



Lecture 22.05: Input Transformations: Scaling & Shifts

Python 3.10+

Numpy Numerical Computing

Status Completed

Language: English | [हिंदी / Hinglish](#) | [Back to Lecture 22](#)



1. Input Argument Transformations

When arithmetic operations are applied **inside** the function's argument:

$$y = f(c \cdot x + d)$$

they transform the graph **horizontally** along the x-axis. Notice that horizontal transformations act counter-intuitively compared to vertical transformations.

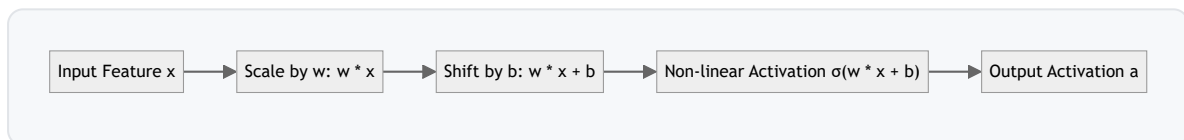
Transformation	Equation	Geometric Effect	Counter-Intuitive Rule
Horizontal Shift Right	$y = f(x - d)$ with $d > 0$	Shifts the curve right by d units	Minus sign shifts right
Horizontal Shift Left	$y = f(x + d)$ with $d > 0$	Shifts the curve left by d units	Plus sign shifts left
Horizontal Compression	$y = f(c \cdot x)$ with $c > 1$	Compresses curve toward y-axis by $\frac{1}{c}$	Multiplying by $c > 1$ shrinks width
Horizontal Stretch	$y = f(c \cdot x)$ with $0 < c < 1$	Stretches curve away from y-axis by $\frac{1}{c}$	Dividing by c widens curve
Horizontal Reflection	$y = f(-x)$	Reflects curve across the y-axis	Inverts horizontal direction

2. The Affine Input Transformation in Neural Networks

In deep learning, every neuron's pre-activation value is an **affine input transformation**:

$$z = w \cdot x + b$$

1. **Weight w (Input Scaling)**: Stretches or compresses the input feature space. In Sigmoid activations $\sigma(wx + b)$, a large w steepens the transition, acting like a step function.
2. **Bias b (Input Shift)**: Shifts the decision threshold left or right, determining the activation baseline.



Feature Standardization as an Input Map:

Data normalization maps inputs via an affine scaling and shifting:

$$\tilde{x} = \frac{x - \mu}{\sigma} = \left(\frac{1}{\sigma}\right) x - \left(\frac{\mu}{\sigma}\right)$$

3. Python Implementation: Visualizing Horizontal Shifts & Scaling

```
import numpy as np
import matplotlib.pyplot as plt

x = np.linspace(-6, 6, 400)
base = x**2

plt.figure(figsize=(9, 5))
plt.plot(x, base, 'k-', lw=2, label='Base: f(x) = x^2')
plt.plot(x, (x - 2)**2, 'r--', lw=2, label='Shift Right: f(x - 2)')
plt.plot(x, (x + 2)**2, 'b-.', lw=2, label='Shift Left: f(x + 2)')
plt.plot(x, (2 * x)**2, 'g:', lw=2, label='Horizontal Compression: f(2x)')
plt.plot(x, (0.5 * x)**2, 'm--', lw=2, label='Horizontal Stretch: f(0.5x)')

plt.axhline(0, color='gray', lw=1)
plt.axvline(0, color='gray', lw=1)
plt.title('Horizontal Input Transformations on f(x) = x^2', fontsize=12, fontweight='bold')
plt.xlabel('x')
plt.ylabel('y')
plt.grid(True, linestyle=':', alpha=0.6)
plt.legend()
plt.ylim(0, 15)
plt.show()
```